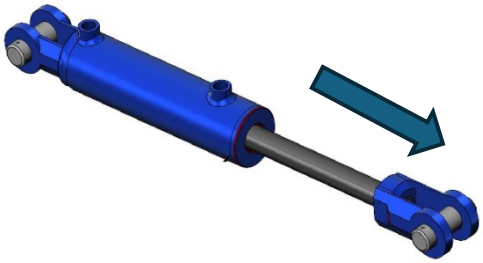
N5 Pneumatics Worksheet



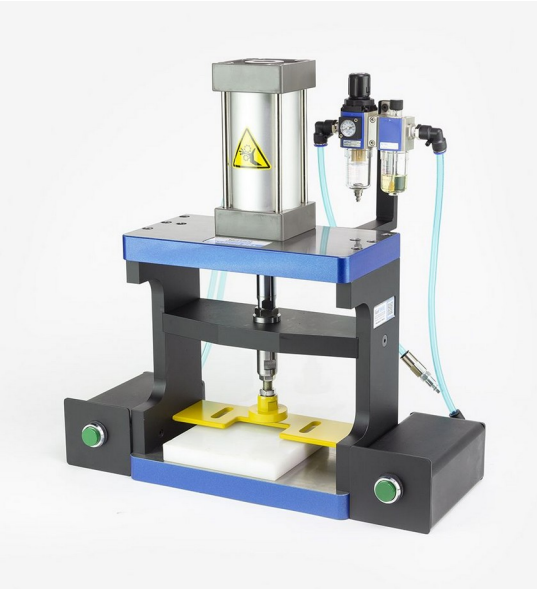
1. Write down the formula that we use to calculate the force in a single-acting cylinder as it outstrokes.



2. A pneumatic stamping machine is used to stamp the company logo onto metal casings. It is discovered that the stamp does not imprint the logo properly. Suggest ways of increasing the size of the force produced by the cylinder.



3. What controls the instroke of a single-acting cylinder?



4. A single-acting cylinder is used to press two sheets of acrylic together when they are gluing. The process requires a force of 300 N. The only piston available has a diameter of 20 mm and it is supplied with air at a pressure of 0.3 N/mm^2. Will this arrangement enable this process to be carried out properly?

5. A double-acting cylinder has a piston diameter of 25 mm and a rod diameter of 10 mm. Pressure = 0.4 N/mm^2. Calculate the instroke force.

6. What force will be produced by a 20 mm diameter cylinder as it goes positive using a pressure of 0.8 N/mm^2?

7. Calculate the outstroke force produced by a 40 mm diameter cylinder when it operates with a supply pressure of 3 bars.

8. Write down the formula that we would use to calculate the pressure of a system if we already knew the force required and the size of the cylinder available.

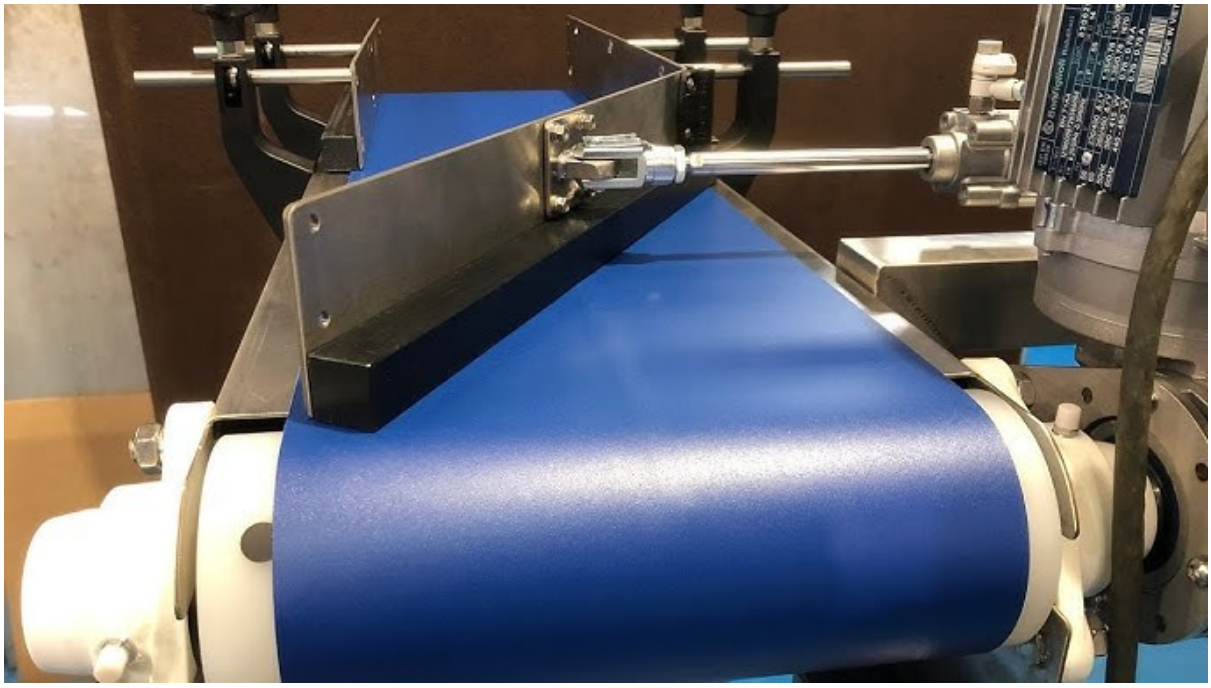
9. A double-acting cylinder has a piston diameter of 60 mm and a rod diameter of 10 mm. The air supply pressure is 12bar. Calculate the instroke force.



10. A stamping machine exerts a force of 454 N with a piston diameter of 34 mm. Calculate the air pressure required for this operation.



11. A machine that places tops on bottles uses a single-acting cylinder. The process requires a force of 650 N. What air pressure needs to be supplied to the cylinder with a diameter of 56 mm?



12. A force of 540 N is needed to push a packing case off a conveyor belt. The single-acting cylinder used has a diameter of 60 mm. What air pressure should be supplied to the system?



13. A pneumatic system is used to test the quality of drawer guides in kitchen cabinets. A force of 16 N is needed to open the drawer. The single-acting cylinder available has a piston diameter of 10 mm. What air pressure should be supplied?

14. Write down the formula we would use to find the area of a piston if we already knew the size of the force it needed to produce and the air pressure being supplied.



15. A single-acting cylinder is used to lift parcels on to a conveyor. This requires a force of 180N with the system operating at a pressure of 6 bars. Calculate the area of the piston required.



16. A door requires a force of 400 N to slide it open. A single-acting cylinder supplied with a pressure of 5 bars controls the operation. Calculate the diameter of the piston required to produce this force.



17. A furnace door weighs 100N and is lifted by a single-acting cylinder as it outstrokes. Compressed air is supplied at a pressure of 4 bars. Calculate the diameter of the piston required to raise the door.